

TrES-1b

R-1202

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-1_211029 · created 2026-07-31T06:51:39+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2459516.6274 +/- 0.0014 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.172 +/- 0.024
Transit depth (Rp/Rs)^2	2.95 +/- 0.83 [%]
Orbital Inclination (inc)	89.91 +/- 0.83
Airmass coefficient 1 (a1)	1.357 +/- 0.016
Airmass coefficient 2 (a2)	-0.164 +/- 0.012
Scatter in the residuals of the lightcurve fit is	1.18 %
Best Comparison Star	None
Optimal Aperture	4.63
Optimal Annulus	11.24
Transit Duration (day)	0.0924 +/- 0.0019

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.172 ± 0.024 vs 0.1358 (deviation 1.07σ)
Duration fit vs published	0.0924 d vs 0.1042 d (deviation 4.4σ)
Mid-time O–C	6.3 min
Depth SNR	5.1
Residual scatter	1.18 %

Light curve

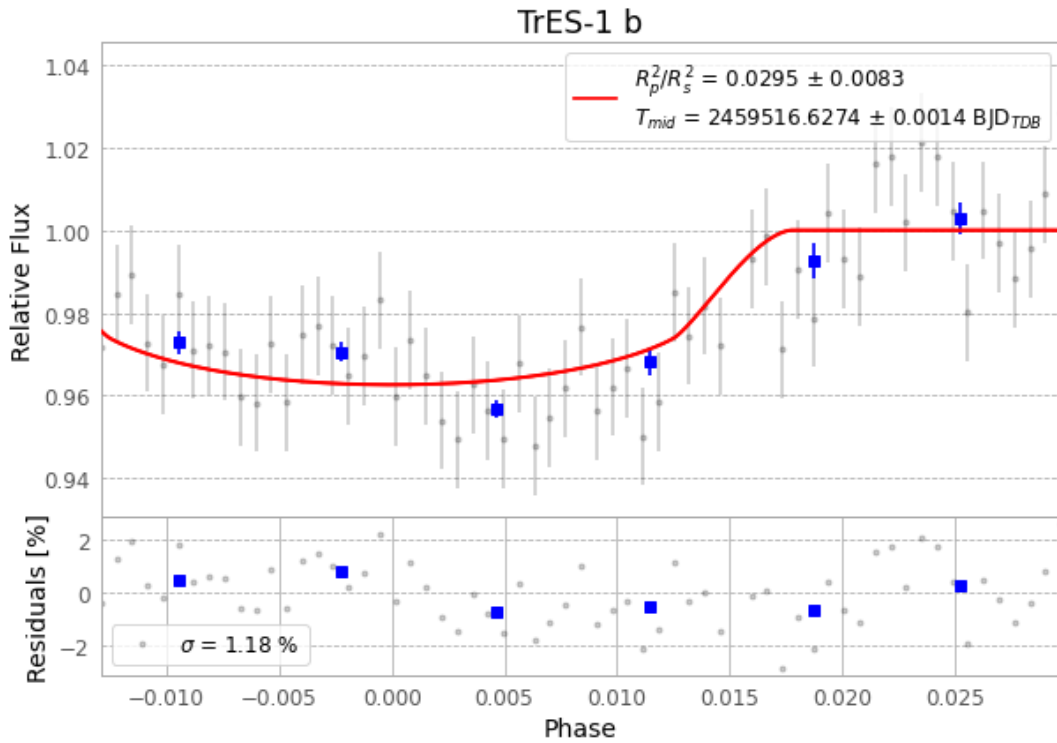


Figure 1 — FinalLightCurve_TrES-1 b_2021-10-28.png (SHA-256 ca9dc20f1f686b70...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [402, 435], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 5485c443fddd487d...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

63 FITS frames (41 MB), TRES-1211029020612.FITS → TRES-1211029051212.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-1-211029.log (d208a963aab2...), FinalLightCurve_TrES-1 b_2021-10-28.png (ca9dc20f1f68...), FinalLightCurve_TrES-1 b_2021-10-28.csv (4a7ff39e8bb2...)

Compute resources

Wall time 120.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-31 06:51Z.